

Radar Signal Monitoring AS IS

Current implementation guide for Radar candidate and signal search

Status: AS IS

Pipeline id: signal-monitoring

Generated PDF: docs/radar/pipelines/signal-monitoring/RADAR_SIGNAL_MONITORING_AS_IS.pdf

Current behavior slice: 0.7.6.4.19.1: Signal monitoring live benchmark reachability and cross-criterion evidence reconciliation

Implemented TO BE: docs/radar/pipelines/signal-monitoring/to-be/RADAR_SIGNAL_MONITORING_TO_BE_0.7.6.4.19.1.md

Validation report: docs/radar/pipelines/signal-monitoring/validation/0.7.6.4.18.2.2/VALIDATION_REPORT.md

Configuration validation: docs/radar/pipelines/signal-monitoring/validation/0.7.6.4.18.3.1/VALIDATION_REPORT.md

Product-surface validation: docs/radar/pipelines/validation/0.7.6.4.18.3.2/VALIDATION_REPORT.md

Reproducibility validation: docs/radar/pipelines/signal-monitoring/validation/0.7.6.4.19.1/VALIDATION_REPORT.md

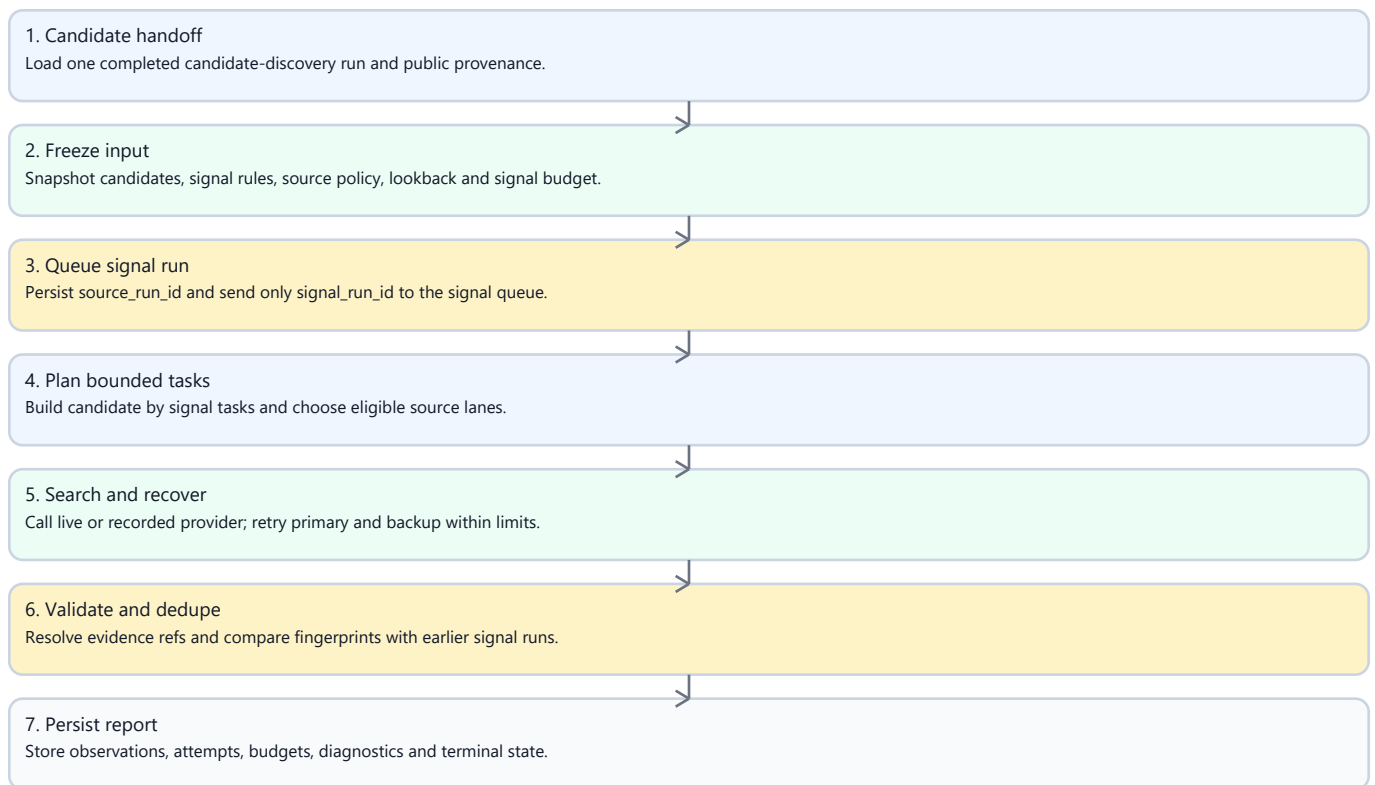
1. Purpose And Boundary

Signal Monitoring checks already discovered candidates for time-bounded intent evidence. It consumes an immutable, evidence-complete candidate-discovery run, but it does not rediscover companies, expand the candidate universe, call the candidate-discovery executor or spend candidate-discovery budgets.

The pipeline supports deterministic recorded execution and persisted live execution through its own API, Celery job, provider adapter, output repository and budget report. A signal run is linked to its source candidate run by `source_run_id`; candidate history continues to select only `pipeline_id=candidate_discovery`.

2. Implemented Flow

Figure 1. End-to-end Radar execution flow



3. Input And Source Planning

SignalMonitoringInputAssembler accepts only a completed candidate run from the same Radar. The default scope contains evidence-complete accepted and review-needed candidates; accepted_only and explicit candidate/signal whitelists are supported. The input snapshot freezes candidate provenance, signal rules, source policy, the common as_of, previous fingerprints, previous source keys and per-lane watermarks.

For each candidate x criterion, the deterministic planner considers:

- up to two concrete known-source URLs;
- an official-domain task when policy permits it;
- configured criterion-specific sources;
- a separate open-web task when policy permits it.

Known-source tasks carry URL, title, snippet and source identity. Registry refs without a URL are not treated as fresh-signal sources. Official tasks contain real domain restrictions. Every selected source decision appears in the lane ledger as scheduled, executed, not_scheduled_budget_limited, not_executable or policy_limited; selected decisions cannot silently disappear. These are the implemented contracts SM-PLAN-01, SM-SRC-01, SM-SRC-02 and SM-SRC-03.

4. Retrieval Audit And Source Lifecycle

Every executed task produces a product-safe `SignalSearchExecutionReceipt` with query, candidate, criterion, lane, URL/domain constraints, engine, effective window, timestamps, result count, normalized source refs and terminal outcome (SM-AUD-01). Provider annotations and citations are sanitized and normalized; credentials, headers, raw payloads and hidden reasoning are never persisted.

The source lifecycle records planned, requested, retrieved, verified, linked, used, no-results, rejected and failed transitions. This makes a searched negative result auditable: the report shows where the pipeline searched, under which restrictions and what came back.

4.1 Product-Facing Cumulative Surface

The persisted report remains the immutable result of one signal run. The application read model in `signal_monitoring/surface.py` projects a cumulative product view for one selected run and its predecessors that share the same Radar and source candidate run.

The view keeps current-run delta separate from cumulative state, resolves product-safe evidence from normalized report collections, and records the originating signal run. It presents candidate count, criterion count and candidate-criterion check count separately. A check is not a found signal.

The same selected surface is rendered in the report, candidate list and candidate detail. Candidate identity and qualification still come exclusively from candidate discovery; candidates outside monitoring scope are explicitly marked not monitored.

5. Evidence Validation And Checkpoints

A positive signal is accepted only when all checks pass:

- the evidence belongs to the requested candidate;
- it supports the requested criterion;
- its date is within the effective window;
- every used ref resolves to normalized evidence;
- policy and source capability permit the evidence;
- known-source evidence belongs to the requested URL family;
- generated summary and score do not contradict the evidence.

Irrelevant group-company pages, another legal entity's evidence, out-of-window documents and unresolved refs cannot become observed (SM-VAL-01, SM-VAL-02). One bounded query revision is allowed per candidate/criterion; primary and backup retries remain bounded and visible.

After required lanes finish, the checkpoint produces `observed`, `not_observed`, `bounded revision`, `review_needed_coverage_incomplete`, `not_searched_budget_limited` or `provider_or_schema_recovery_needed`.

not_observed is legal only when all required lanes have successful receipts and no valid evidence (SM-OBS-01). Failure or missing coverage is never reported as proof that no signal exists.

not_observed never means "the pipeline did not search". It means the required search coverage is proven by successful receipts and yielded no valid signal.

5.1 Event-Time Integrity And Source Capability

Signal evidence separates retrieval time, publication time and event time:

- retrieved_at records when the pipeline obtained a page and never proves freshness;
- published_at records a publication date when it can be established from safe source metadata, URL/title/snippet hints or provider extraction;
- event_at / event_end_at records the event interval when the source text supports it.

Temporal statuses are confirmed_in_window, review_needed_date_unknown, review_needed_date_conflict and rejected_out_of_window. Relevant evidence with no reliable date is retained for human review. Known out-of-window evidence is retained as rejected evidence but cannot be a confirmed source. When one task has both confirmed and rejected evidence, only confirmed refs are published in source_refs; rejected or unknown evidence remains in the artifact for audit/control evaluation.

Every source carries a generic capability such as official_press, generic_web, project_or_asset_history, identity_only, registry or unknown. Identity-only and registry sources may support candidate identity, but they cannot confirm a fresh signal. Candidate/source binding is checked before known-source scheduling so another entity's page cannot become a candidate-specific signal source.

5.2 Criterion-Owned Search And Cross-Criterion Reconciliation

Signal criteria own their search and evidence vocabulary. Planning combines candidate legal name and aliases with configured criterion terms and bounded query variants; production logic does not branch on benchmark criterion codes, company names, control URLs or a hardcoded calendar year (SM-QUERY-02).

Provider-local source references are task-scoped before they enter the shared artifact, preventing collisions such as unrelated tasks both returning source_1. Product-safe evidence retrieved for one criterion may be tested against another enabled criterion for the same candidate. Reuse requires a new entity, criterion, temporal, capability and source-binding decision, while retaining the originating task, criterion, receipt and source reference (SM-XCRIT-01). Evidence is never copied across candidates or accepted only by a broad keyword (SM-XCRIT-02). Capability is enforced per evidence ref, so a

mixed response cannot use an `identity_only` or `registry` ref to confirm a fresh signal (SM-CAP-03).

Canonical URL identity removes fragments and tracking parameters but preserves host and path identity. This identity is shared by evidence validation, incremental source keys and post-run control evaluation (SM-URL-01).

6. Time Windows And Incremental Runs

Initial lookback precedence is:

1. explicit API run override;
2. criterion `initial_lookback_days` when present;
3. Radar `monitoring_policy.lookback_window`;
4. 365-day fallback.

The CLI passes no implicit seven-day override. The benchmark Radar therefore uses its configured 90 days unless a run explicitly requests 365 days.

SM-WIN-01 covers this precedence.

Without an explicit override, a repeated run resolves an independent watermark for `candidate_id` + `signal_code` + `source_lane`. Its window begins at the last successful `searched_through_at` minus a two-day overlap and ends at the new `as_of` (SM-WIN-02). Only successful searched receipts advance that lane.

Provider errors, schema failures, policy skips and budget limits do not advance it (SM-WIN-03).

Overlap dedupe uses both exact fingerprints and stable `candidate` + `criterion` + `canonical evidence URL` keys. Different provider wording therefore cannot republish an old event as new (SM-DED-01).

6.1 Effective Per-Signal Policy

Every signal criterion owns an additive `monitoring_policy`: enabled state, initial lookback, incremental overlap, cadence and enabled source lanes. Initial window precedence is explicit run override, criterion policy, Radar policy and finally the visible 365-day system default. Incremental overlap is resolved for the same candidate, criterion and lane as its watermark. The planner intersects criterion lanes with global source policy and never schedules a disabled lane.

Preflight and persisted artifacts expose the effective values and their basis.

Cadence is configuration only; this slice does not create automatic jobs. These contracts implement SM-CFG-01, SM-CFG-02, SM-CFG-03 and SM-WIN-04.

6.2 Monitoring Series And Reproducibility

An optional `monitoring_series_id` isolates fingerprints, source keys and watermarks for reproducibility testing without deleting persisted history.

Independent initial series A and B start empty; incremental C loads only the B

series (SM-REP-02, SM-DED-03). Normal product runs retain the default series behavior.

The accepted reproducibility contour is explicit rather than hidden in an aggregate observed count. Each independent initial run must find at least three of four frozen positive controls, one must find all four, and their union must find all four. Every run still passes negative, unknown-date, temporal, source, provenance, receipt and budget integrity checks (SM-REP-03). A single approved exact-URL miss is retained as provider_search_drift (SM-DRIFT-01); it is not represented as a successful match. The original two-times-4/4 v1 manifest and machine FAIL remain archived alongside the approved v2 amendment (SM-REP-01).

7. Budgets And Models

Signal Monitoring owns independent counters and never reads or mutates candidate-discovery budgets.

signal_monitoring_smoke

Tasks	6
Provider calls	8
Extraction retries	2
Backup retries	1
Lookback queries	6
Source verifications	12
Query revisions	bounded

signal_monitoring_quality

Tasks	48
Provider calls	60
Extraction retries	8
Backup retries	4
Lookback queries	60
Source verifications	120
Query revisions	1 per candidate/criterion

The live adapter uses signal_monitoring_default. Transport and schema errors become explicit review/recovery states, not false searched-negative results.

8. Persisted Artifact And API

radar_signal_monitoring_report stores run lineage, immutable input, model profile, search plan, plan acceptance, source-lane ledger, receipts, source lifecycle, window policy, watermarks before/after, validation records, checkpoint decisions, observations, provider attempts, signal-only budgets and diagnostics. It survives API and worker restarts.

The API remains additive and backward compatible:

text

GET /api/radars/{radar_id}/signal-monitoring/preflight

POST /api/radars/{radar_id}/signal-monitoring-runs

GET /api/radars/{radar_id}/signal-monitoring-runs

GET /api/signal-monitoring-runs/{run_id}

GET /api/signal-monitoring-runs/{run_id}/report

Radar configuration loading is resource-oriented. GET /api/radars is a summary catalog; GET /api/radars/{radar_id} returns the active definition and no history by default; the 20-run history contains only display metadata and scalar output counts. Full run metadata and immutable configuration snapshots are direct-run resources. The frontend requests definition/history only after one Radar is opened, candidates only for the selected run, and dossier, journal, trace and full signal report only on explicit interaction. This is the implemented RADAR-API-01 and RADAR-LAZY-01 contract.

Settings distinguish active definition from a selected run snapshot. Loading, failure, stale response and dirty local draft are explicit states; local demo overrides do not silently replace backend definitions in API mode. Docker UI acceptance opened the benchmark Radar ten cold times at 1280x720 and 1366x768, always showing 2 qualification rules, 3 signals and 3 sources (RADAR-UI-01, RADAR-UI-02). The complete traceability gate is SM-PROC-03.

Recorded contract execution remains available through:

powershell

python -m power_web_os.demo run-recorded-signal-monitoring

Persisted live execution uses run-live-signal-monitoring with the API at <http://127.0.0.1:8001>, a completed source candidate run, candidate ids, signal codes and either the smoke or quality profile.

9. Validation Evidence

Recorded positive-control benchmark contains known events for industrial candidates and two criteria. It requires exact evidence matching, zero accepted irrelevant controls and complete lane/receipt accounting.

Persisted live acceptance used source candidate run
radar-run-3bbf9c0f-330e-4468-8901-966a751234a8:

- initial quality run signal-run-010ef75d-c626-44e3-a025-56c95522c1a8:

six candidates, two criteria, 12 candidate/criterion pairs, 27 executed tasks, four observed outcomes, two unknown-date review items, four of four positive controls matched by candidate/criterion/URL-set/date, two negative controls retained unconfirmed, zero retrieved-at freshness violations and zero confirmed out-of-window evidence;

- incremental quality run signal-run-df00b3b8-267c-4091-a4dd-8167434e2cf3:

six candidates, 25 executed tasks, 24 loaded previous source keys, 25 incremental windows, three duplicate review items, zero previous sources republished and zero observed outcomes republished as new.

Both reports remained readable through the persisted API contour. The machine validation report is the closure authority, not these prose numbers.

Slice 0.7.6.4.19.1 added a separate reproducibility chain over source candidate run radar-run-b03fac86-7307-448f-8deb-c1ea1794956c:

- independent A signal-run-8eb6d673-a6f6-417a-8519-8cc50e7e94f8 found 4/4 positive, 4/4 negative and 1/1 unknown-date controls;
- independent B signal-run-6754eba6-a43a-4594-8236-d7ed60f6d2c5 found 3/4 positive, 4/4 negative and 1/1 unknown-date controls; its missing frozen Khimprom URL is recorded as provider search drift;
- incremental C signal-run-47e29772-8cbf-421e-8072-7c2d951ba611 loaded 67 B-series source keys, suppressed two confirmed and three review duplicates, republished zero previous evidence, had zero receipt gaps and zero illegal watermark advances.

All three reports remained byte-stable and readable after API and worker restart. Exact-URL search stability remains planned in 0.7.6.4.19.2; a provider-independent fallback is conditional in 0.7.6.4.19.3.

10. Requirement Change Record

Slice 0.7.6.4.18.2.1 finalized the following mandatory requirements in this AS IS: SM-PLAN-01, SM-SRC-01, SM-SRC-02, SM-SRC-03, SM-AUD-01, SM-OBS-01, SM-WIN-01, SM-WIN-02, SM-WIN-03, SM-VAL-01, SM-VAL-02, SM-DED-01, SM-ARCH-01 and SM-PROC-01.

Slice 0.7.6.4.18.2.2 finalized the following mandatory requirements in this AS IS: SM-TIME-01, SM-TIME-02, SM-TIME-03, SM-CAP-01, SM-BIND-01, SM-BIND-02, SM-QUERY-01, SM-RETRY-01, SM-SCORE-01, SM-BENCH-01, SM-BENCH-02, SM-BENCH-03, SM-DED-02, SM-AUD-02, SM-ARCH-02 and SM-PROC-02.

Slice 0.7.6.4.19.1 finalized SM-REP-01, SM-REP-02, SM-REP-03, SM-DRIFT-01, SM-QUERY-02, SM-XCRIT-01, SM-XCRIT-02, SM-CAP-03, SM-URL-01, SM-DED-03 and SM-PROC-03. The approved deviation and original v1 FAIL are preserved under the slice validation directory.

The traceability chain is TO BE -> acceptance manifest -> exact tests -> two persisted live reports -> validation JSON/Markdown -> this AS IS (SM-PROC-01, SM-PROC-02, SM-PROC-03). Signal Monitoring remains isolated from candidate-discovery internals (SM-ARCH-01, SM-ARCH-02).

11. Process Rule

Every future Radar slice marked Pipeline and Behavior change: true requires an AS IS baseline, TO BE Markdown/PDF, acceptance manifest, mapped tests, diagnostic run, bounded autofix/RCA loop, persisted validation report with validation_status=PASS, and final TO BE-to-AS IS reconciliation. The roadmap tracker refuses Done when this evidence is missing or failed.

12. Product UI Contract

The Radar Runs tab exposes Signal Monitoring through its persisted API. It has its own preflight, run action, history selector, status, budget summary and report. Every selected signal run displays source_run_id and automatically selects that candidate-discovery run. Candidate history and counters remain unchanged by signal runs. The full cross-pipeline contract lives in docs/radar/pipelines/RADAR_PIPELINE_SPLIT_UI_CONTRACT.md.

13. Out Of Scope

- recurring scheduler and notification delivery;
- persisted/UI per-criterion depth, cadence and source settings, planned in 0.7.6.4.18.3.1;
- candidate rediscovery or candidate-universe expansion;
- public quality claims from one acceptance pair.